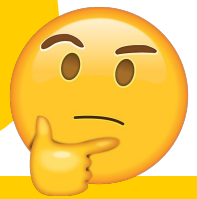
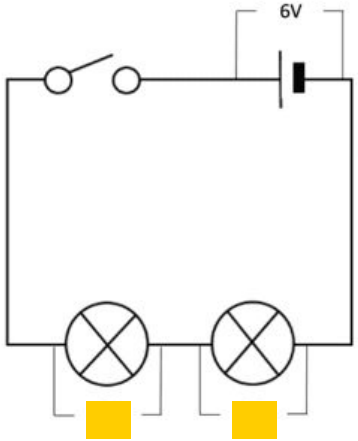
Voltage in parallel circuits

Lesson objectives

- To draw circuit diagrams of parallel circuits.
- To describe the rule for voltages in a parallel circuit.

Brain teaser





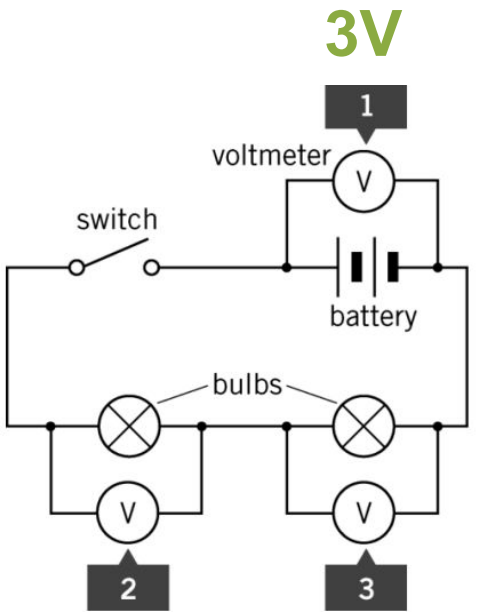
Key words: voltage, potential difference, voltmeter, volts, parallel circuit

Potential difference in a series circuit

When two or more components are connected in **series**, **the total potential difference of the supply is shared between them**.

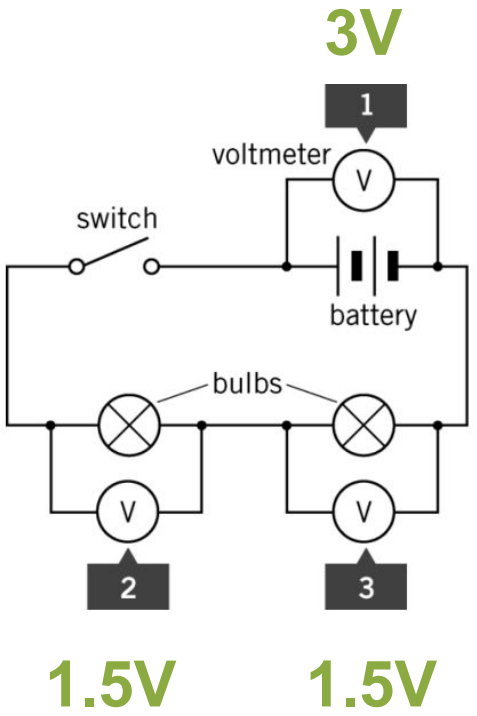
This means that if you add together the voltages across each component connected in series, the total equals the voltage of the power supply.

E.g. 3V = 1.5V + 1.5V

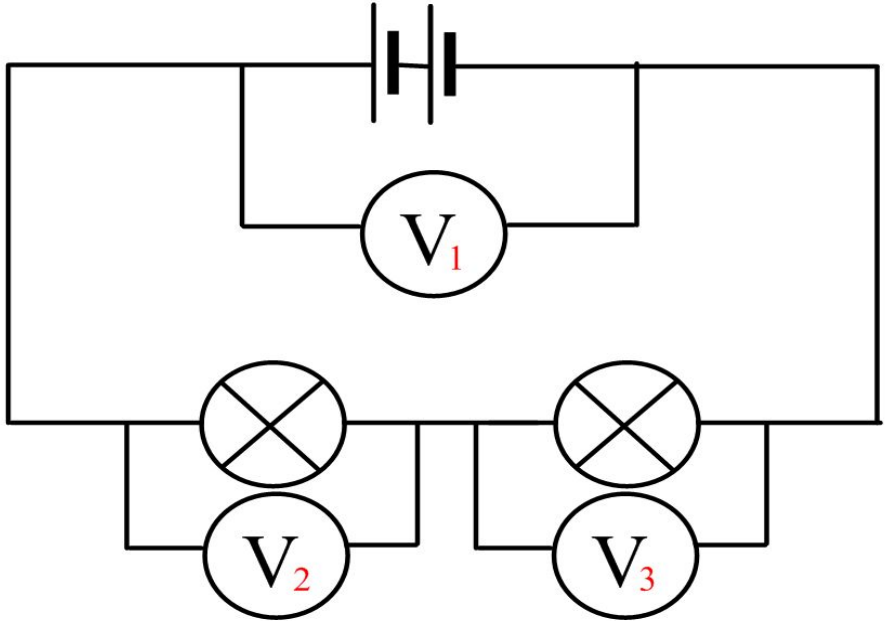


Potential difference in a series circuit

The **more cells**, the **larger voltage/potential difference** which will **transfer more energy** to the bulbs.

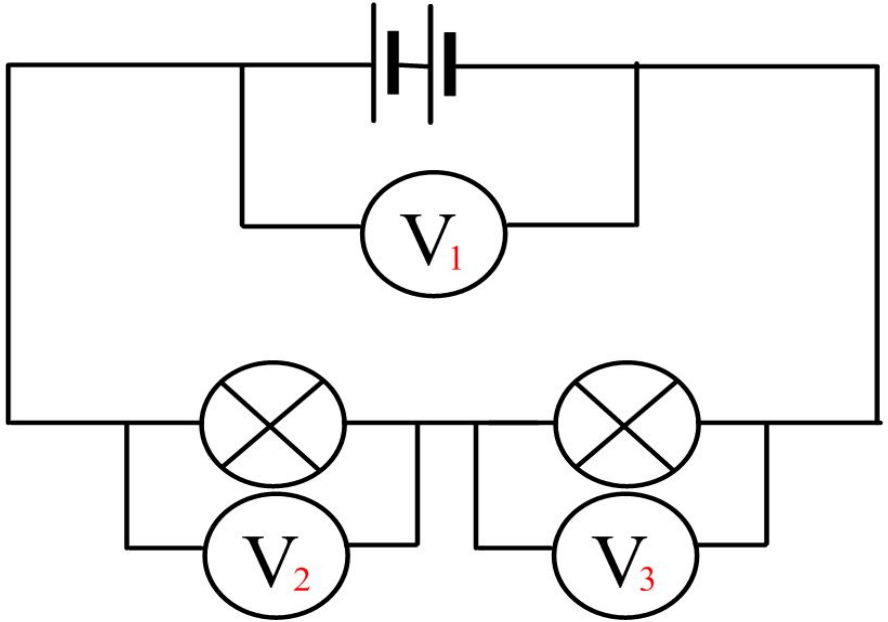


Concept check! *Series Circuit*



State the readings on V_1 and V_3 if V_2 reads 3.2V (assume the bulbs are identical).

Concept check! *Series Circuit*



V_1 is 6.4V

V_3 is 3.2V

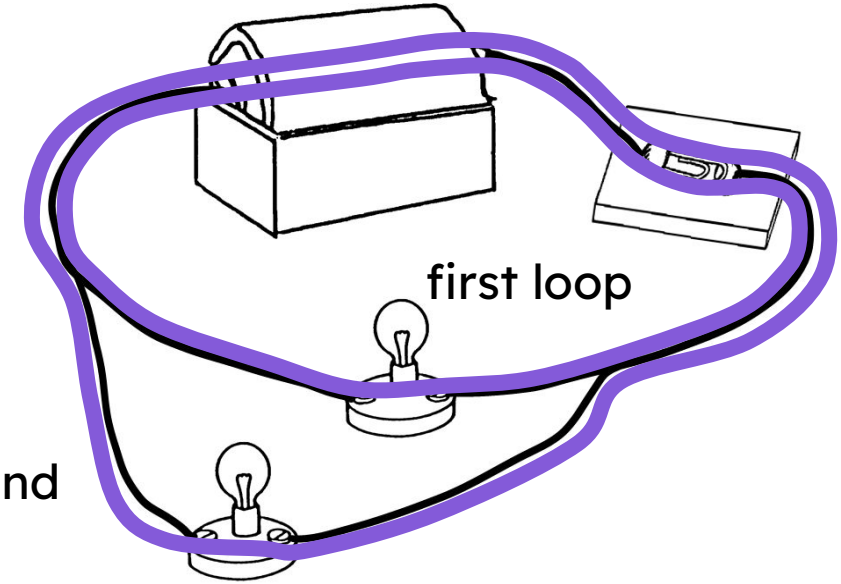
- in a parallel circuit the voltage across each component adds up to the voltage across the battery.

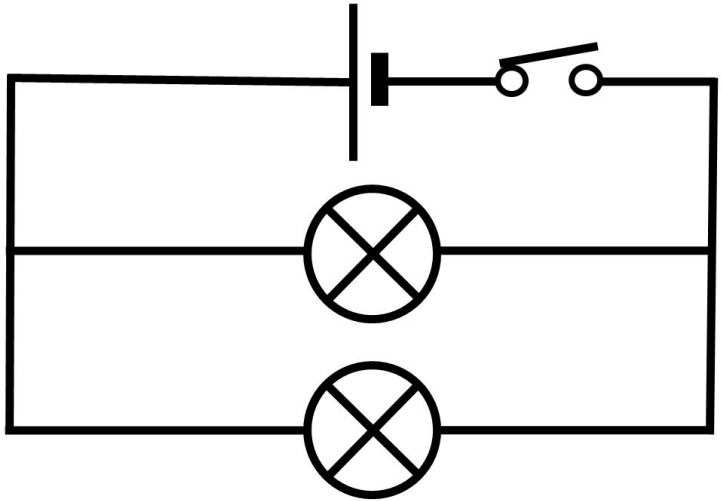
Parallel circuits

Explanation

This is a **parallel circuit**.

There is more than one loop from the cell.

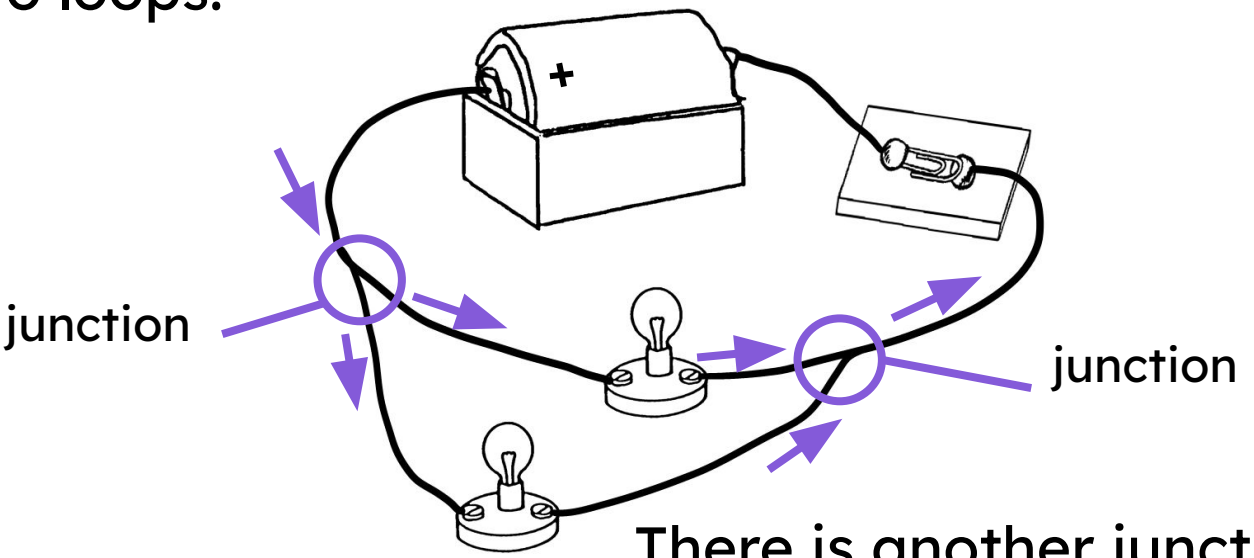




Parallel circuits

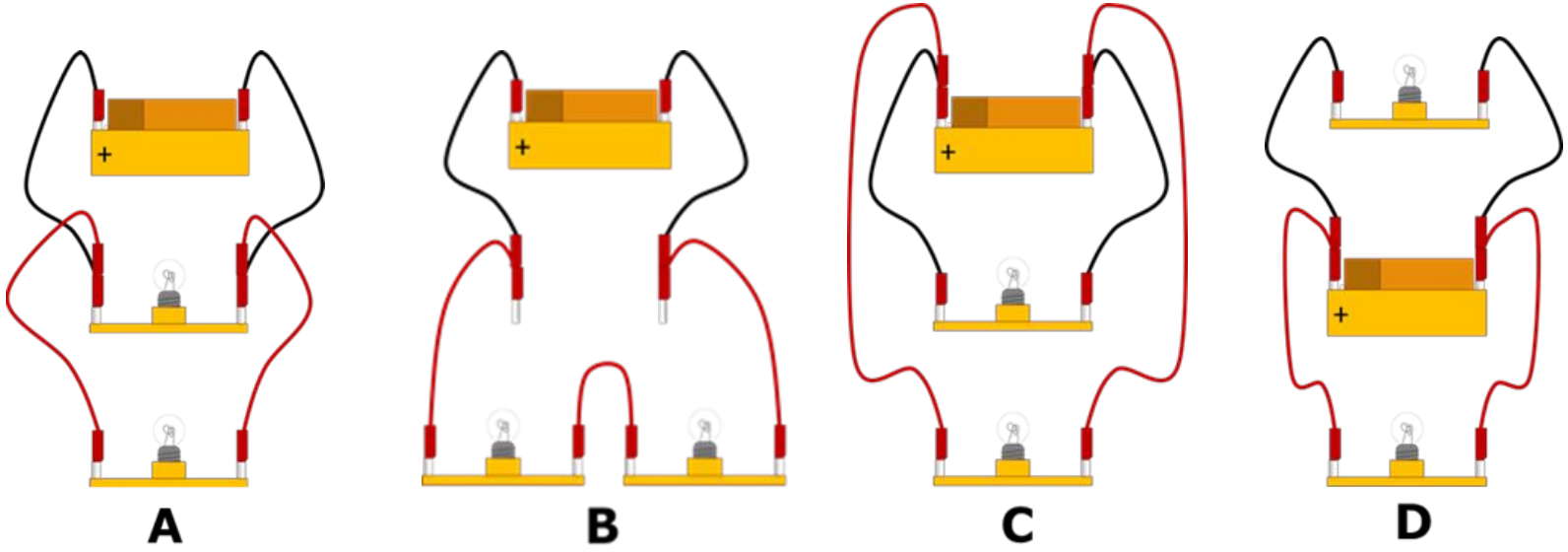
Explanation

There is a junction where the current splits into the two loops.



There is another junction where the current from the two loops joins up.

Identify which of the circuits (A, B, C and D) work the same way. Explain your answer.



Identify which of the circuits (A, B, C and D) work the same way. Explain your answer.

Circuits A, C and D are the same parallel circuits, as they have junctions leading to two separate loops.

The two loops in each circuit each have one lamp, so they are the same.

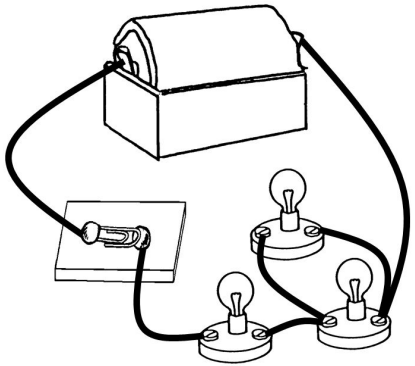
Therefore, circuits A, C and D will work the same way, even though the lamps and wire positions are different.

Circuit B is a series circuit, which is different from the others.

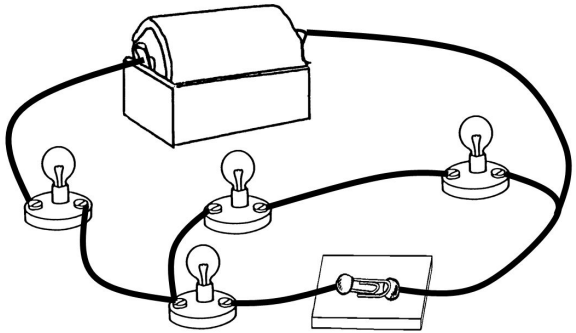
Drawing parallel circuit diagrams

Draw circuit diagrams for the circuits below.

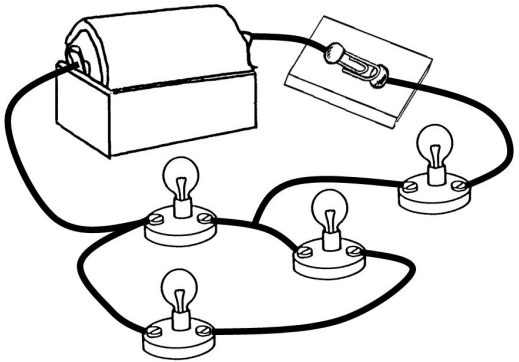
1)



2)



3)

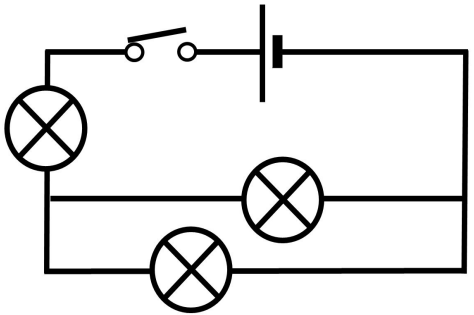


Remember:

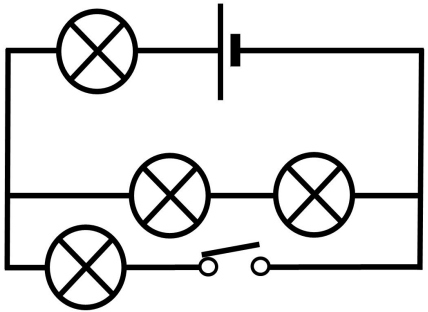
- ❏ Circuit diagrams should be drawn one loop at a time.
- ❏ Make sure there are no tiny gaps in the lines, especially at corners and junctions.
- ❏ Horizontal and vertical lines for wires should be drawn with a ruler.

Drawing parallel circuit diagrams

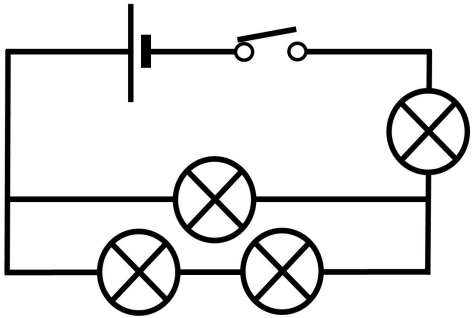
1)



2)



3)



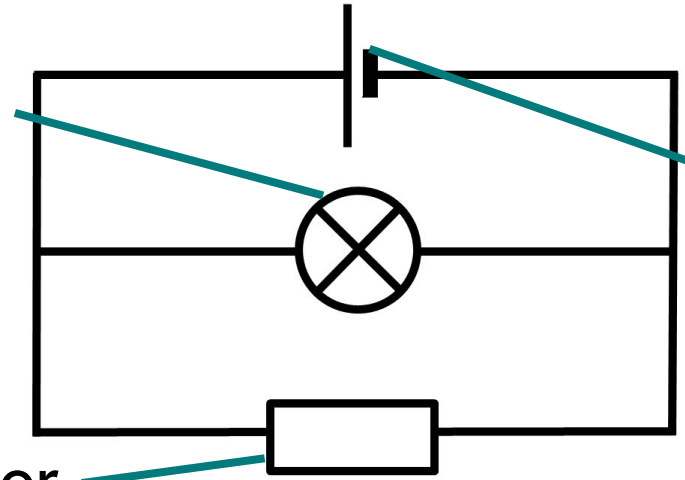
Your diagrams should be similar to these.

A rule for voltage in parallel circuits

Explanation

Voltage across parallel branches.

The voltage across the lamp was 1.4 V.



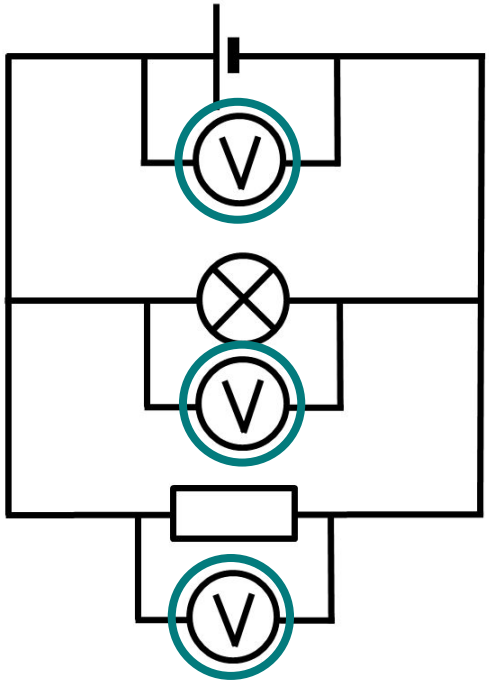
The voltage across the cell is also 1.4 V.

The voltage across the resistor was also 1.4 V.

A rule for voltage in parallel circuits

Explanation

The three **voltmeters** in the circuit below will all read the same **voltage**.



The voltage across the cell is the same as...

...the voltage across each of the two parallel branches.

A rule for voltage in parallel circuits

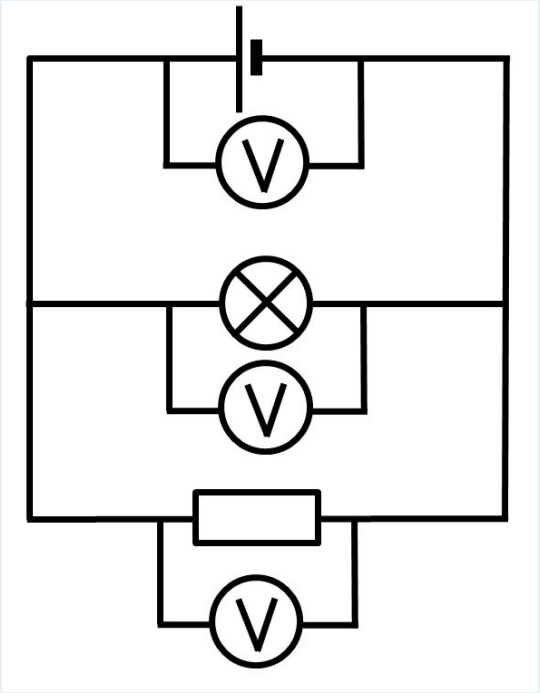
Check

What are the voltages across the lamp and the resistor in the circuit below? The cell has a voltage of 2.0 V.

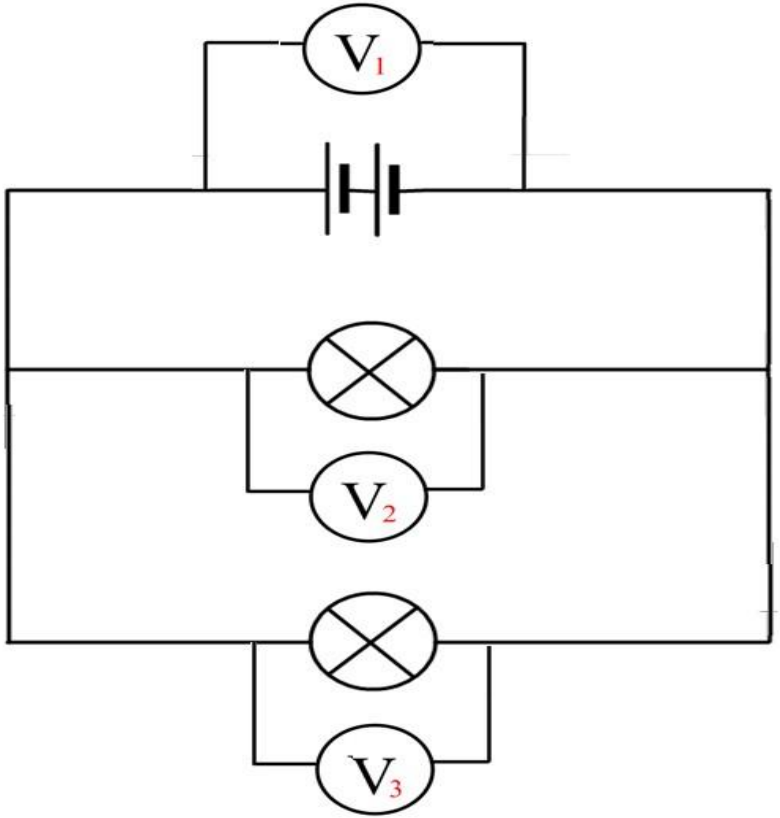
a lamp = 1.0 V, resistor = 0.0 V

b lamp = 1.0 V, resistor = 1.0 V

c lamp = 2.0 V, resistor = 2.0 V ✓

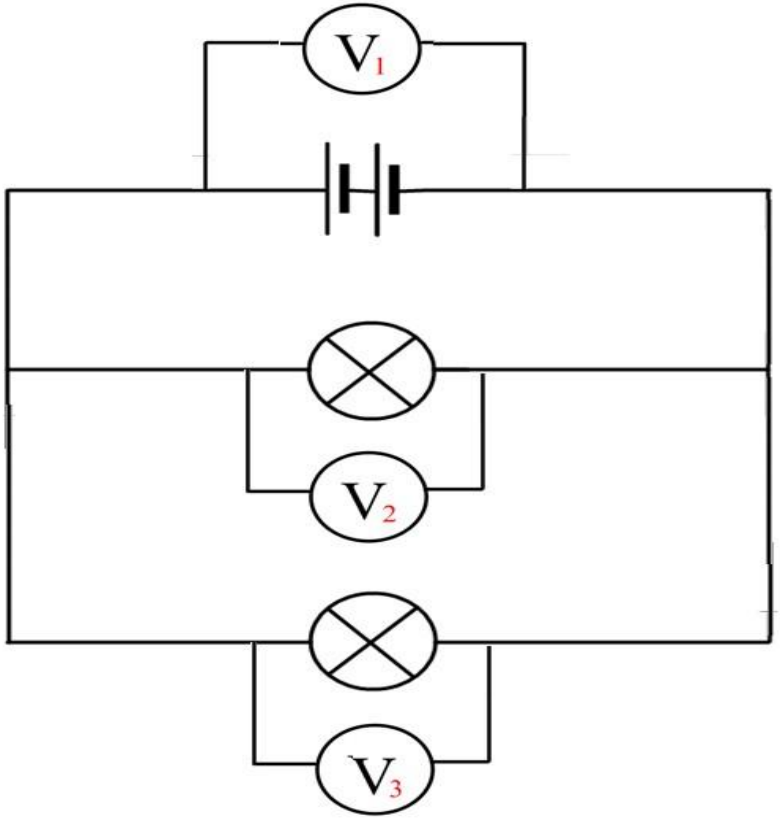


Concept check! *Parallel Circuit*



State the readings on V_1 and V_2 if V_3 reads 1.4V.

Concept check! *Parallel Circuit*



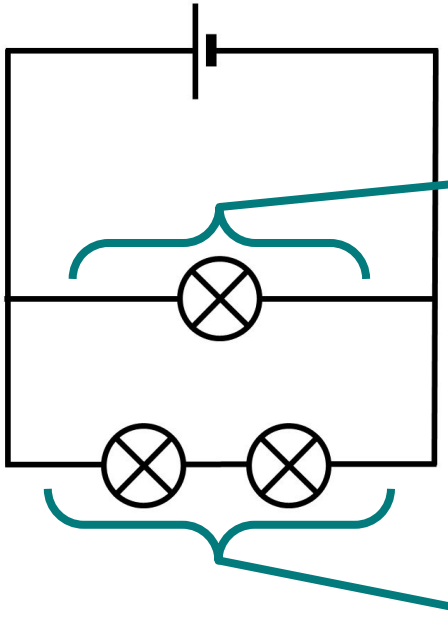
V_1 and V_2 read 1.4V

– in a parallel circuit the voltage across each branch is the same.

A rule for voltage in parallel circuits

Explanation

The lamps in the **parallel circuit** below are identical.



The **voltage** across the branch with one lamp...

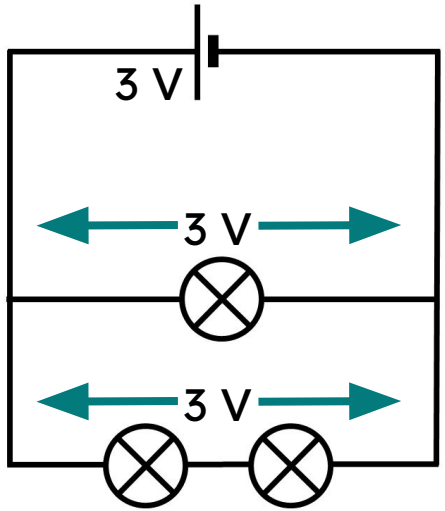
...is the same as the voltage across the branch with two lamps.

A rule for voltage in parallel circuits

Explanation

The rule for **voltage** in a **parallel circuit** is:

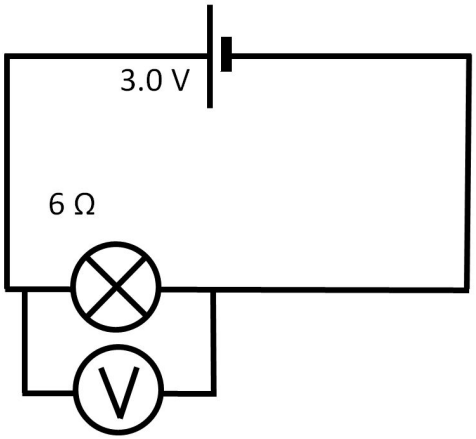
The voltage across each branch of a parallel circuit is the same as the voltage across the cell or battery.



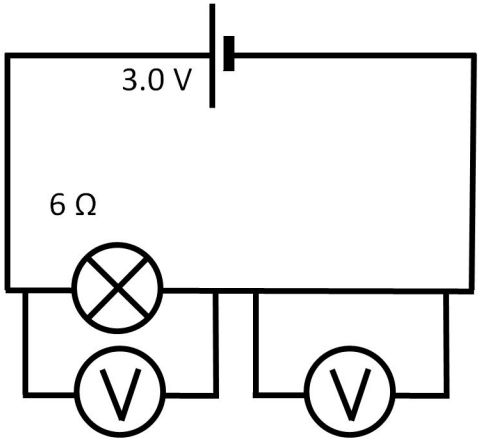
A rule for voltage in parallel circuits

Explanation

A **voltmeter** needs to be connected across a component with resistance to measure a **voltage**.



3.0 V



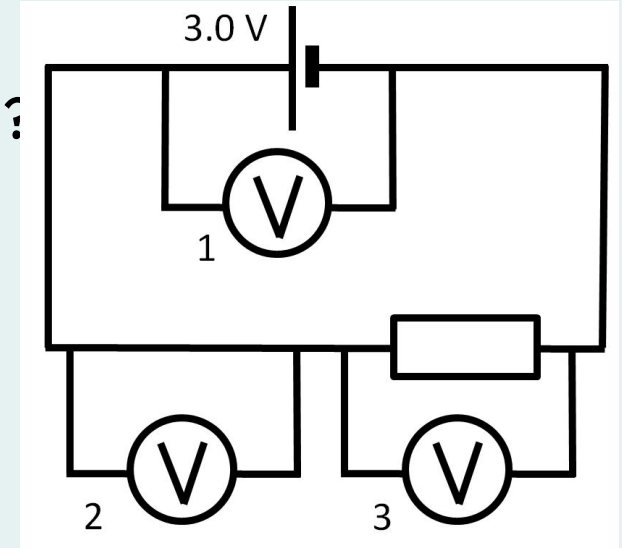
3.0 V 0.0 V

The leads have zero resistance so have a voltage of zero across them.

A rule for voltage in parallel circuits

Check

What are the correct readings for voltmeters 1, 2 and 3 in the circuit shown?



a 1 = 3.0 V, 2 = 3.0 V and 3 = 3.0 V

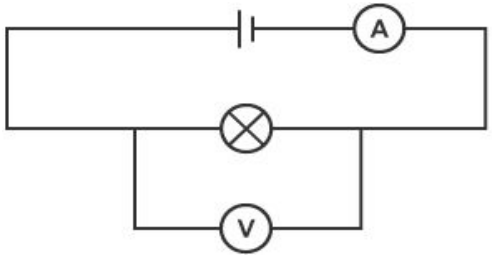
b 1 = 3.0 V, 2 = 0.0 V and 3 = 3.0 V ✓

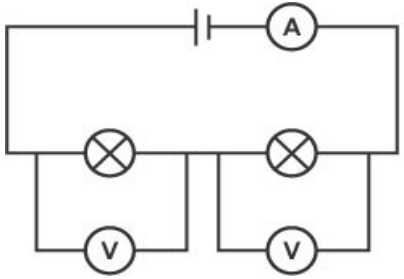
c 1 = 3.0 V, 2 = 1.0 V and 3 = 2.0 V

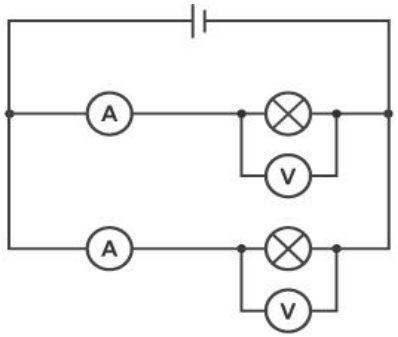
Investigate the brightness of bulbs in series and parallel circuits

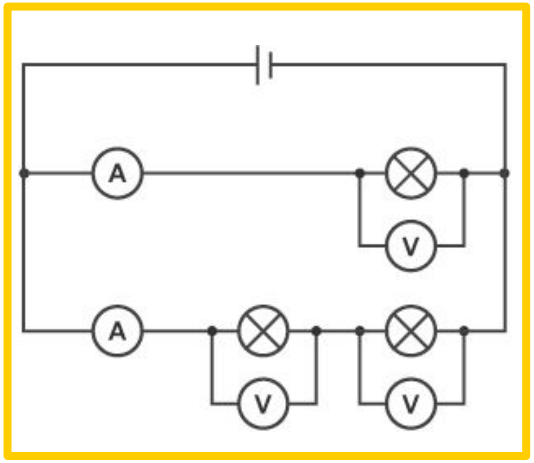
Aim of the experiment

To see how the brightness of bulbs change as they are connected in series and parallel, and to measure the potential difference across the bulbs and the current through the bulbs in series and parallel.



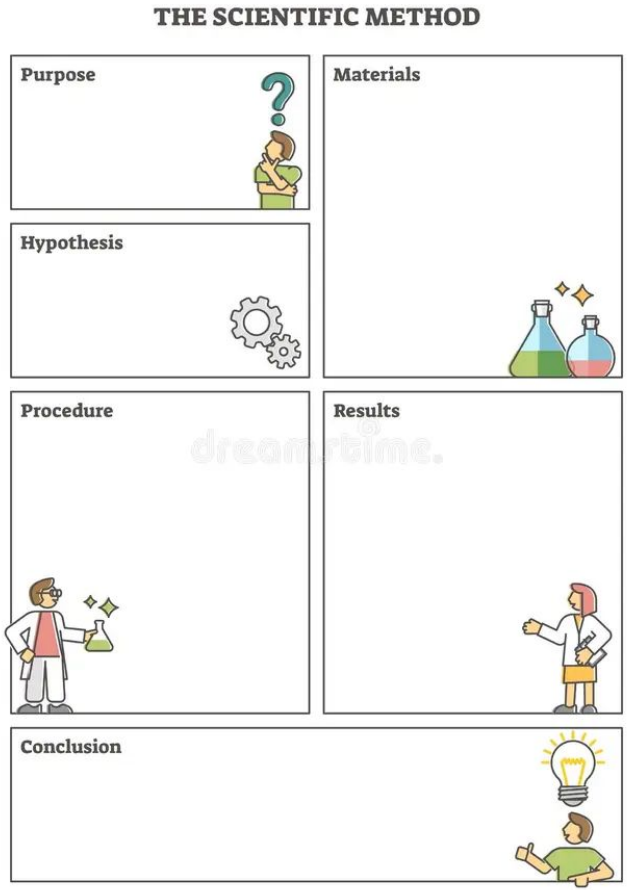






Optional challenge circuit

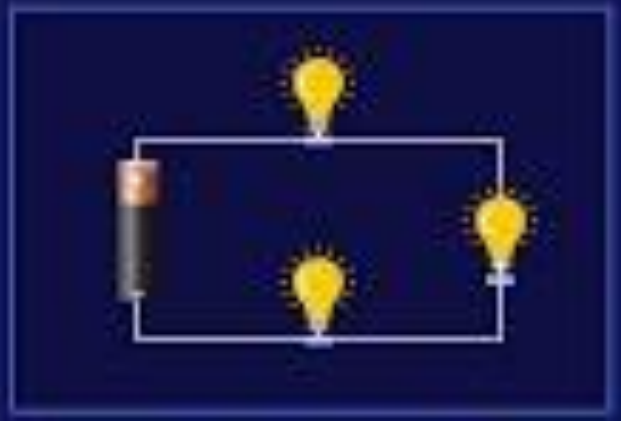
Investigate the brightness of bulbs in series and parallel circuits



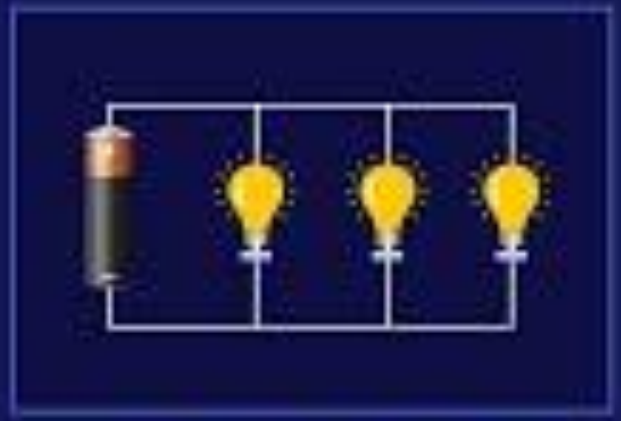
✅ **Success Criteria**

Section	What You Need to Include	Mandarin Support (中文帮助)
Title	Clear, bold title that explains what the experiment is about.	实验标题 – 简明扼要地说明实验内容
Aim	Write the aim in your own words. E.g. "To compare brightness and measure current & voltage in series and parallel."	实验目的 – 比较串联和并联中灯泡亮度、电流和电压。
Circuit Diagrams	Neat diagrams for both series and parallel circuits. Label components clearly.	电路图 – 包括清晰标注的串联电路和并联电路。
Method	Use bullet points or numbers. Keep it simple and in order. Include safety.	实验步骤 – 使用项目符号或编号。步骤要清晰有序，注意安全。
Photos	Include clear photos of you doing the experiment and of the circuits.	照片 – 包括你做实验和电路的清晰照片。
Results	Use a table to show measurements of current (A) and potential difference (V). Mention bulb brightness.	实验结果 – 用表格展示电流（A）和电压（V）的数据，写出灯泡亮度的变化。
Analysis / What you found out	Write what happened in both types of circuit. Compare the results. Why was one brighter?	分析 – 比较两个电路的结果。哪个更亮？为什么？
Conclusion	A short paragraph summarising what you found.	结论 – 总结你学到了什么。
Presentation	Neat, colourful, easy to read. Titles and subtitles. Correct spelling.	展示 – 整洁、有颜色、易读。有标题和副标题。拼写要正确。

Brightness of Bulbs in Series and Parallel Circuits



Series Circuit



Parallel Circuit